



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Test Plan (STP)

Blood Bank Management System

Version 1.0 (Draft)

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Document Information

Field	Value
Project	Blood Bank Management System (BBMS)
Document	Software Test Plan (IEEE 829 structure)
Version	1.0
Authors	Team 9 —Purav N Khincha (PES1UG24CS349) (Testing & QA Lead), with module inputs from Members 1, 2 and 4
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Revision History

Name	Date	Reason for Changes	Version
Purav N Khincha (PES1UG24CS349)	03-10-2026	Initial test plan with test cases and SRS traceability	1.0

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1. Introduction

1.1 Purpose

This document defines the test plan for the Blood Bank Management System (BBMS) Version 1.0. It describes the objectives, scope, strategy, environment, schedule, responsibilities and test cases used to verify that the system satisfies the functional, non-functional, safety and security requirements in the BBMS Software Requirements Specification (SRS) v1.0.

1.2 Scope

Testing covers all features specified in the SRS: donor registration and management, donation recording, blood inventory management, expiry and stock monitoring, blood availability search, blood request management, blood issue and fulfilment, reports and history, authentication and role-based access control. Testing is performed at unit, integration, system and acceptance levels.

Medical screening of donors, laboratory testing of blood, physical storage hardware and integrations with external hospital systems are outside the scope of this plan (see Section 4).

1.3 References

- [1] Blood Bank Management System — Software Requirements Specification, v1.0, Team 9, 04-09-2026.
- [2] Blood Bank Management System — Software Architecture and Design Specification (SAD), v1.0, Team 9.
- [3] IEEE Std 829-2008, Standard for Software and System Test Documentation.
- [4] IEEE Std 29148-2018, Requirements Engineering.
- [5] OWASP Application Security Verification Standard (ASVS) 4.0 and OWASP Top 10.
- [6] Django Software Foundation, Django Documentation — Testing and Deployment Checklist.

1.4 Definitions and Acronyms

Term	Meaning
BBMS	Blood Bank Management System
SRS / SAD / STP	Software Requirements Specification / Software Architecture & Design Specification / Software Test Plan
RTM	Requirements Traceability Matrix
FR / NFR	Functional Requirement / Non-Functional Requirement
SEC / SO / SAFE / BR / DR	Security Requirement / Security Objective / Safety Requirement / Business Rule / Domain Requirement (SRS IDs)

Unit (blood)	A single bag of donated blood, the counting unit for inventory
Near-expiry	A blood unit expiring within the configured monitoring period (default 7 days)
Low stock	A blood group whose available units fall below the minimum level (default 5 units)
UAT	User Acceptance Testing
P95	95th-percentile response time

2. Test Items

The following software items of BBMS v1.0 will be tested. Module names correspond to the components defined in the SAD.

Test Item (Module)	Description
Authentication & User Management	Login/logout, role assignment (Blood Bank Staff, Hospital Staff, Administrator), user administration
Donor Management	Donor registration, search, update, donation history
Donation Recording	Recording donations and their status against registered donors
Blood Inventory	Stock per blood group, additions on accepted donation, deductions on issue
Expiry & Stock Monitoring	Detection of expired, near-expiry units and low-stock blood groups
Blood Availability Search	Search of available units by blood group
Blood Request Management	Request submission, validation, status tracking
Blood Issue & Fulfilment	Approval/rejection, issue against approved requests, inventory update
Reports & History	Donation, request, issue and inventory reports
Web UI and REST/API endpoints	Browser-based screens and the endpoints defined in SAD Section 4.3

3. Features to be Tested

Each feature is mapped to the requirement IDs in the SRS. Section 13 gives the complete mapping from requirement to test case.

Feature	SRS Requirement IDs	Test Cases
Donor Registration & Management (5.1)	FR-1.1 - FR-1.7	TC-DON-01 to 03
Donation Recording (5.2)	FR-2.1 - FR-2.7, BR-08	TC-DNT-01, 02
Blood Inventory Management (5.3)	FR-3.1 - FR-3.5, BR-02, BR-07	TC-INV-01, TC-ISS-01, 03
Expiry & Stock Monitoring (5.4)	FR-4.1 - FR-4.5, BR-03	TC-MON-01, 02
Blood Availability Search (5.5)	FR-5.1 - FR-5.4	TC-SRCH-01, 02
Blood Request Management (5.6)	FR-6.1 - FR-6.7, DR-04	TC-REQ-01, 02
Blood Issue & Fulfilment (5.7)	FR-7.1 - FR-7.7, BR-04 - BR-06	TC-ISS-01 to 03
Reports & History (5.8)	FR-8.1 - FR-8.5	TC-RPT-01
Performance (6.1)	NFR-01 - NFR-04	TC-PERF-01 to 03, TC-INV-01
Usability, Reliability, Portability (6.4)	NFR-05, NFR-06, NFR-08	TC-USE-01, TC-REL-01, TC-PORT-01
Safety (6.2)	SAFE-01 - SAFE-04	TC-MON-01, TC-ISS-01 to 03
Security (6.3)	SEC-01 - SEC-06, SO-01 - SO-03	TC-SEC-01 to 06

4. Features Not to be Tested

Feature / Item	Reason
Medical screening and donor/blood suitability (SAFE-05)	Performed by authorized blood-bank personnel outside the software (SRS 2.6, SAFE-05)
Laboratory testing of donated blood	Physical procedure outside the system boundary
Integration with external hospital systems, live location services, third-party blood databases	Not part of Version 1.0 (SRS 2.5, 2.6)
E-mail / SMS notifications	Not required in Version 1.0 (SRS 3.3)
Donor self-service portal and eligibility check	Future scope (SRS 7.2)
Internals of Django, the RDBMS, OS and web browsers	Third-party components assumed to be tested by their vendors
NFR-07 Maintainability	Verified by code/design review, not by executable test cases

5. Test Approach / Strategy

Test Levels

Level	Scope and Method	Tools / Owner
Unit	Models, forms, validators and service functions of each Django app tested in isolation (e.g., stock never negative, expiry calculation, request status transitions). Target $\geq 80\%$ statement coverage per app.	Django TestCase, pytest-django, coverage.py — each module owner
Integration	Interaction between modules and with the database: donation $\rightarrow$ inventory, request $\rightarrow$ approval $\rightarrow$ issue $\rightarrow$ inventory, role checks on views and endpoints.	Django test client, Postman — module owners with QA Lead
System	End-to-end workflows through the browser on the staging build, plus performance, security and portability tests.	Selenium, Locust, OWASP ZAP — QA Lead
Acceptance (UAT)	Representative users (Staff, Hospital Staff, Administrator roles played by team members/faculty) execute key scenarios against acceptance criteria in the SRS.	Manual — whole team, sign-off by Project Lead

Test Types

- **Functional testing:** positive, negative and boundary-value cases for every FR (equivalence partitioning on blood groups, units and dates; boundary values for near-expiry window and minimum stock).
- **Regression testing:** the automated unit and integration suites run on every pull request through the CI pipeline; the full suite runs before each sprint release.
- **Performance testing:** load tests with 20 concurrent users on a database seeded with 10,000 donors (the SRS "normal operating conditions") to verify NFR-01 to NFR-03.
- **Reliability testing:** fault injection to confirm transactional rollback (NFR-06).
- **Usability testing:** timed task completion with first-time users (NFR-05).
- **Compatibility testing:** Chrome, Edge and Firefox on Windows, macOS and Linux (NFR-08).
- **Security testing:** see Section 5.1.

Entry Criteria

- SRS v1.0 and SAD v1.0 baselined and available in the team repository.
- Build deployed to the test environment and passes the smoke test (login, search, request).

- Test data seeded (donors, units across all 8 blood groups, users for each role).
- Unit tests for the module under test pass in CI.

Exit Criteria

- 100% of planned test cases executed; $\geq 95\%$ passed.
- 0 open Critical and 0 open High severity defects; Medium/Low defects documented with an agreed plan.
- 100% of functional requirements traced to at least one passed test case (RTM complete).
- All SAFE and SEC test cases passed.
- Performance targets NFR-01 to NFR-03 met.

5.1 Security Validation

Security validation verifies the security objectives (SO-01 Confidentiality, SO-02 Integrity, SO-03 Access Control) and security requirements SEC-01 to SEC-06 from SRS Section 6.3, and the threats identified in the SAD security architecture.

Requirement	Validation Activity	Test Case / Tool
SEC-01	Access every protected URL and endpoint without a session; verify redirect to login or 401. Verify session invalidation on logout.	TC-SEC-01; manual + Django test client
SEC-02, SEC-03	Role-permission matrix test: each role attempts each protected action through the UI and through direct HTTP requests (to catch hidden-button-only protection).	TC-SEC-02; Postman collection
SEC-04	Non-admin roles attempt user creation, role change and deactivation.	TC-SEC-03
SEC-05	Verify masked password fields and salted-hash storage (no plain-text passwords in DB or logs).	TC-SEC-04; DB inspection
SEC-06	Verify HTTP→HTTPS redirect, TLS 1.2+, Secure/HttpOnly cookies and HSTS header on the deployed build.	TC-SEC-05; browser dev tools, SSL Labs
SO-01, SO-02	Input fuzzing of search, donor and request fields with SQL-injection and XSS payloads; CSRF token check on all POST forms.	TC-SEC-06; OWASP ZAP baseline scan
SO-02	Confirm that inventory changes occur only through approved transactions and no partial updates remain after failure.	TC-REL-01, TC-ISS-02

All	Run Django's deployment check (manage.py check --deploy) and static security analysis (Bandit) in CI; no High findings allowed at release.	CI pipeline (Member 4)
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6. Test Environment

Component	Details
Hardware	Developer laptops (8 GB RAM minimum) for unit/integration tests; one staging server/VM (2 vCPU, 4 GB RAM) for system, performance and security tests
Server software	Python 3.12, Django 5.x, relational DBMS as finalised in the SAD (PostgreSQL assumed; SQLite for local unit tests), Gunicorn + Nginx with TLS on staging
Client software	Latest Google Chrome, Microsoft Edge and Mozilla Firefox on Windows 11, macOS and Ubuntu Linux
Test tools	pytest-django / Django TestCase, coverage.py, Selenium WebDriver, Postman, Locust (load), OWASP ZAP, Bandit
Management tools	GitHub (repository, pull requests), GitHub Actions (CI), Jira (test cases and defect tracking) — as set up by Member 4
Test data	Fixture/seed script creating synthetic donors (no real personal data), blood units across all 8 groups including expired and near-expiry units, requests in each status, and one account per role. A large seed (10,000 donors, 30,000 units) is used for performance tests.

7. Test Schedule

The schedule is aligned to the project sprints and may be revised by the Project Lead.

Milestone	Start	End
Test plan and test case design	03-10-2026	07-10-2026
Test environment and CI setup	08-10-2026	12-10-2026
Unit and integration testing (with Sprint 1 & 2 development)	08-10-2026	28-10-2026
System testing (functional, security, performance)	29-10-2026	06-11-2026
Regression testing and defect retest	07-11-2026	10-11-2026
User acceptance testing	11-11-2026	13-11-2026
Test summary report	14-11-2026	14-11-2026

8. Test Deliverables

- Software Test Plan (this document)
- Test case specifications (Section 15) and the Jira test case repository
- Automated test scripts (unit, integration, Selenium, Locust) in the project repository
- Test data seed scripts and fixtures
- Test execution logs and CI reports, including coverage reports
- Defect reports (Jira)
- Updated Requirements Traceability Matrix (SRS Appendix C)
- Test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Purav N Khincha (PES1UG24CS349)	Prepare and maintain the test plan; design system, performance and security test cases; coordinate execution; maintain RTM test columns; report metrics. Unit/integration tests for Blood Search and Request Management.
Test Engineer (Donor & Donation)	Prakruti Prasanna Bhat (PES1UG24CS330)	Unit and integration tests for Donor and Donation modules; verify requirement wording supports testable cases.
Test Engineer (Inventory & Monitoring)	Pallavi Rajkumar Bubanale (PES1UG24CS314)	Unit and integration tests for Inventory and Expiry/Stock Monitoring; confirm test items match SAD components.
Test Engineer (Issue & Reports) / CI	Ramith Munnann Ravindranath (PES1UG24CS367)	Unit and integration tests for Blood Issue and Reports; configure CI to run tests, coverage and static analysis on each pull request.
Developers	All members	Fix defects in owned modules and support triage.
Product Owner / Sign-off	Ramith Munnann Ravindranath (PES1UG24CS367), Faculty guide	Approve test results and release readiness.

10. Risks and Mitigation

Risk	Mitigation
Implementation of a module is delayed, blocking integration tests	Write unit tests alongside code; use stubs for dependent modules; prioritise High-priority FRs
SRS requirement IDs or thresholds change (e.g., NFR IDs, near-expiry window)	Keep RTM in one place; update affected test cases in the same pull request as the SRS change
Small team means the same person develops and tests a module	Cross-review: each system test case is executed by a member who did not build that module
Staging server unavailable or under-powered for load tests	Run performance tests locally with a documented configuration as fallback; record environment details with results
Insufficient realistic test data	Seed script generates synthetic data for all blood groups and edge cases (expired, near-expiry, zero stock)

11. Assumptions & Dependencies

- The SRS v1.0 requirements (including NFR-01 to NFR-09 and the default thresholds of 7 days near-expiry and 5 units minimum stock) are baselined before system testing.
- The SAD defines the module/component names and API endpoints used in test items and test cases.
- A staging environment with HTTPS is available for system, performance and security tests.
- GitHub Actions CI and Jira are configured by Member 4 before integration testing starts.
- Only synthetic test data is used; no real donor information is stored in any test environment.

12. Suspension & Resumption Criteria

Suspend testing if

- The test environment or database is unavailable for more than 4 hours.
- A build blocks more than 30% of planned test cases (e.g., login or database migration failure).
- A Critical defect corrupts inventory data, making further results unreliable.

Resume testing when

- The blocking defect is fixed and verified in a new build.
- The environment is restored and the smoke test passes.
- Test data has been re-seeded to a known state.

13. Test Case Management & Traceability

Test cases are identified as TC-<MODULE>-<NN> (DON = Donor, DNT = Donation, INV = Inventory, MON = Monitoring, SRCH = Search, REQ = Request, ISS = Issue, RPT = Reports, PERF = Performance, REL = Reliability, USE = Usability, PORT = Portability, SEC = Security). Each test case records the SRS requirement IDs it verifies. The matrix below traces every SRS requirement to its test cases and will be copied into the Test Case ID column of SRS Appendix C.

SRS Requirement ID	Requirement (brief)	Test Case ID(s)
FR-1.1 - FR-1.3	Donor registration, mandatory fields, blood group	TC-DON-01, TC-DON-02
FR-1.4, FR-1.5	Search and update donor	TC-DON-03
FR-1.6	Donor donation history	TC-DNT-01
FR-1.7	Donor validation messages	TC-DON-02
FR-2.1 - FR-2.6	Record donation details and history	TC-DNT-01
FR-2.7	Reject invalid donation	TC-DNT-02
FR-3.1, FR-3.2, FR-3.4	Maintain / add / view inventory	TC-INV-01
FR-3.3	Reduce inventory on issue	TC-ISS-01
FR-3.5	Inventory never negative	TC-ISS-03
FR-4.1, FR-4.2, FR-4.5	Expiry tracking and exclusion	TC-MON-01
FR-4.3, FR-4.4	Near-expiry and low-stock detection	TC-MON-02
FR-5.1, FR-5.2	Search by blood group, show quantity	TC-SRCH-01

FR-5.3	Exclude expired units from search	TC-SRCH-01, TC-MON-01
FR-5.4	Indicate unavailable group	TC-SRCH-02
FR-6.1 - FR-6.6	Submit, review and track request	TC-REQ-01
FR-6.7	Reject incomplete request	TC-REQ-02
FR-7.1	Approve / reject request	TC-ISS-01, TC-ISS-02
FR-7.2	Issue only against approved request	TC-ISS-02
FR-7.3, FR-7.4	Verify stock before issue	TC-ISS-03
FR-7.5 - FR-7.7	Update inventory, status, issue record	TC-ISS-01
FR-8.1 - FR-8.5	History and reports	TC-RPT-01
NFR-01	Search / inventory view ≤ 3 s	TC-PERF-01
NFR-02	Submissions ≤ 3 s	TC-PERF-02
NFR-03	Reports ≤ 5 s	TC-PERF-03
NFR-04	Inventory reflected immediately	TC-INV-01
NFR-05	Record donation within 2 min	TC-USE-01
NFR-06	No partial inventory updates	TC-REL-01
NFR-07	Modular Django apps	Verified by code/design review (no test case)
NFR-08	Browser / OS portability	TC-PORT-01
NFR-09	Every FR has a test case	Verified by this RTM (100% FR coverage)
SEC-01	Authentication required	TC-SEC-01
SEC-02, SEC-03	Role-based access; staff-only inventory/issue	TC-SEC-02
SEC-04	Admin-only user management	TC-SEC-03
SEC-05	Password masking	TC-SEC-04
SEC-06	HTTPS	TC-SEC-05
SAFE-01	Expired units not available	TC-MON-01
SAFE-02	Issue only on approved request	TC-ISS-02
SAFE-03	No issue on insufficient stock	TC-ISS-03
SAFE-04	Confirmation before issue	TC-ISS-01
SAFE-05	Medical screening outside software	Not tested (out of scope, Section 4)

Coverage summary: all 47 functional requirements (FR-1.1 to FR-8.5) are covered by at least one test case. NFR-01 to NFR-06 and NFR-08 are covered by test cases; NFR-07 and NFR-09 are verified by review. All six security requirements and safety requirements SAFE-01 to SAFE-04 are covered.

14. Test Metrics & Reporting

Metric	Definition	Target
Test execution %	Executed test cases ÷ planned test cases	100%
Pass %	Passed ÷ executed	≥ 95%
Requirement coverage	FRs with ≥ 1 passed test case ÷ total FRs	100%
Code coverage	Statement coverage from coverage.py per Django app	≥ 80%
Defect density	Defects ÷ KLOC per module	Tracked per module
Defect aging	Days a defect stays open, by severity	Critical ≤ 1 day, High ≤ 3 days

Reports: CI test results on every pull request; a weekly test status summary during the sprint (executed, passed, failed, blocked, open defects by severity); a final Test Summary Report at the end of UAT.

Defect severity levels: Critical (data corruption, security breach, system unusable), High (core feature fails, no workaround), Medium (feature fails with workaround), Low (cosmetic).

15. Test Case Specifications

This section specifies 28 test cases covering functional requirements, non-functional requirements, safety and security. Actual Result and Status are filled during execution.

15.1 Test Case Summary

ID	Title	Type	Priority
TC-DON-01	Register a donor with valid details	Functional	High
TC-DON-02	Reject donor registration with missing / invalid data	Functional (Negative)	High
TC-DON-03	Search and update an existing donor	Functional	Medium
TC-DNT-01	Record a donation for a registered donor	Functional	High
TC-DNT-02	Reject donation with invalid data	Functional (Negative)	High

TC-INV-01	Accepted donation updates blood-group inventory	Functional	High
TC-MON-01	Expired units are flagged and excluded from availability	Functional / Safety	High
TC-MON-02	Near-expiry and low-stock detection (boundary values)	Functional (Boundary)	Medium
TC-SRCH-01	Search availability for an available blood group	Functional	High
TC-SRCH-02	Search availability for an unavailable blood group	Functional (Negative)	High
TC-REQ-01	Submit a valid blood request and track its status	Functional	High
TC-REQ-02	Reject a blood request with missing mandatory data	Functional (Negative)	High
TC-ISS-01	Approve a request and issue blood with sufficient stock	Functional / Safety	High
TC-ISS-02	Block issue against a pending or rejected request	Functional / Safety (Negative)	High
TC-ISS-03	Block issue when stock is insufficient	Functional / Safety (Negative)	High
TC-RPT-01	View history and basic reports	Functional	Medium
TC-PERF-01	Search and inventory view response time	Non-functional (Performance)	High
TC-PERF-02	Transaction submission response time	Non-functional (Performance)	Medium
TC-PERF-03	Report generation time	Non-functional (Performance)	Medium
TC-REL-01	Failed issue does not partially update inventory	Non-functional (Reliability)	High
TC-USE-01	Staff can record a donation within 2 minutes	Non-functional (Usability)	Medium
TC-PORT-01	Cross-browser compatibility	Non-functional (Portability)	Low
TC-SEC-01	Unauthenticated access is blocked	Security	High
TC-SEC-02	Role-based access: Hospital Staff cannot modify inventory or issue blood	Security	High
TC-SEC-03	Only Administrator can manage users	Security	High
TC-SEC-04	Password masking and secure storage	Security	High

TC-SEC-05	HTTPS enforced on deployment	Security	Medium
TC-SEC-06	Injection and XSS attempts in search and request fields	Security	High

15.2 Detailed Test Cases

TC-DON-01	Register a donor with valid details
Requirement IDs	FR-1.1, FR-1.2, FR-1.3, DR-01
Type / Priority	Functional / High
Preconditions	Logged in as Blood Bank Staff. No donor exists with the test contact number.
Test Steps	1. Open Donor Registration. 2. Enter all mandatory fields with valid values. 3. Click Save.
Test Data	Name: Ravi Kumar; DOB: 12-05-1998; Gender: M; Blood group: O+; Phone: 9876543210; Address: Bengaluru
Expected Result	Donor record is saved with blood group O+, a unique donor ID is generated, and a success message is shown. Donor appears in donor search.
Actual Result	
Status	Not Executed

TC-DON-02	Reject donor registration with missing / invalid data
Requirement IDs	FR-1.2, FR-1.7
Type / Priority	Functional (Negative) / High
Preconditions	Logged in as Blood Bank Staff.
Test Steps	1. Open Donor Registration. 2. Leave Blood group empty and enter phone as '98AB'. 3. Click Save.
Test Data	Blood group: (blank); Phone: 98AB; other fields valid
Expected Result	Record is NOT saved. Field-level validation messages are shown for Blood group (required) and Phone (invalid format).
Actual Result	
Status	Not Executed

TC-DON-03	Search and update an existing donor
Requirement IDs	FR-1.4, FR-1.5
Type / Priority	Functional / Medium
Preconditions	Donor from TC-DON-01 exists. Logged in as Blood Bank Staff.
Test Steps	1. Search donor by phone 9876543210. 2. Open the donor record. 3. Change Address to 'Mysuru' and click Update.
Test Data	Search key: 9876543210; New address: Mysuru
Expected Result	Search returns exactly the matching donor. After update the record shows Address = Mysuru and a success message is displayed.
Actual Result	
Status	Not Executed

TC-DNT-01	Record a donation for a registered donor
Requirement IDs	FR-2.1-FR-2.6, FR-1.6, BR-08, DR-02, DR-05
Type / Priority	Functional / High
Preconditions	Donor from TC-DON-01 exists. Logged in as Blood Bank Staff.
Test Steps	1. Select the donor. 2. Click Record Donation. 3. Enter donation date, units and status 'Accepted'. 4. Click Save.
Test Data	Date: today; Units: 1; Status: Accepted
Expected Result	Donation record is stored with date, blood group O+, 1 unit and status Accepted, and is linked to the donor. It appears in the donor's donation history.
Actual Result	
Status	Not Executed

TC-DNT-02	Reject donation with invalid data
Requirement IDs	FR-2.7
Type / Priority	Functional (Negative) / High
Preconditions	Logged in as Blood Bank Staff. Donor exists.

Test Steps	1. Open Record Donation for the donor. 2. Enter units = 0 and a future donation date. 3. Click Save.
Test Data	Units: 0; Date: tomorrow
Expected Result	Donation is NOT saved. Validation messages state that units must be ≥ 1 and the date cannot be in the future. Inventory is unchanged.
Actual Result	
Status	Not Executed

TC-INV-01	Accepted donation updates blood-group inventory
Requirement IDs	FR-3.1, FR-3.2, FR-3.4, NFR-04, BR-02
Type / Priority	Functional / High
Preconditions	Inventory for O+ = 4 units. Logged in as Blood Bank Staff.
Test Steps	1. Note O+ stock in Inventory view. 2. Record an Accepted donation of 1 unit (O+). 3. Record a 'Pending' donation of 1 unit (O+). 4. Re-open Inventory view.
Test Data	O+ opening stock: 4
Expected Result	O+ stock shows 5 immediately after the accepted donation. The pending donation is NOT added to available stock.
Actual Result	
Status	Not Executed

TC-MON-01	Expired units are flagged and excluded from availability
Requirement IDs	FR-4.1, FR-4.2, FR-4.5, FR-5.3, SAFE-01, BR-03
Type / Priority	Functional / Safety / High
Preconditions	A- has 2 units: one expired yesterday, one expiring in 20 days.
Test Steps	1. Open Expiry & Stock Monitoring. 2. Open Blood Availability Search and search A-.
Test Data	Unit U1 expiry: today – 1 day; Unit U2 expiry: today + 20 days
Expected Result	U1 is listed as Expired in monitoring. Availability search for A- shows 1 unit (U1 excluded). U1 is not counted when stock is checked for an issue.
Actual Result	

Status	Not Executed
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TC-MON-02	Near-expiry and low-stock detection (boundary values)
Requirement IDs	FR-4.3, FR-4.4
Type / Priority	Functional (Boundary) / Medium
Preconditions	Near-expiry window = 7 days; minimum stock level = 5 units (SRS defaults).
Test Steps	1. Seed units expiring in 6, 7 and 8 days. 2. Set B- stock to 4 units and B+ to 5 units. 3. Open Expiry & Stock Monitoring.
Test Data	Expiry offsets: +6, +7, +8 days; B- = 4, B+ = 5
Expected Result	Units at +6 and +7 days are flagged Near-expiry; +8 is not. B- is flagged Low Stock; B+ (exactly at minimum) is not.
Actual Result	
Status	Not Executed

TC-SRCH-01	Search availability for an available blood group
Requirement IDs	FR-5.1, FR-5.2, FR-5.3
Type / Priority	Functional / High
Preconditions	Logged in as Hospital Staff. O+ has 5 valid units and 1 expired unit.
Test Steps	1. Open Blood Availability Search. 2. Select blood group O+. 3. Click Search.
Test Data	Blood group: O+
Expected Result	System displays O+ as Available with quantity 5 (expired unit not counted).
Actual Result	
Status	Not Executed

TC-SRCH-02	Search availability for an unavailable blood group
Requirement IDs	FR-5.4
Type / Priority	Functional (Negative) / High
Preconditions	Logged in as Hospital Staff. AB- has 0 valid units.

Test Steps	1. Open Blood Availability Search. 2. Select AB-. 3. Click Search.
Test Data	Blood group: AB-
Expected Result	System displays a clear 'AB- is currently unavailable' message with quantity 0. No error page is shown.
Actual Result	
Status	Not Executed

TC-REQ-01	Submit a valid blood request and track its status
Requirement IDs	FR-6.1-FR-6.6, DR-04
Type / Priority	Functional / High
Preconditions	Logged in as Hospital Staff (City Hospital).
Test Steps	1. Open New Blood Request. 2. Enter blood group, units and patient/reason details. 3. Submit. 4. Open My Requests. 5. Log in as Blood Bank Staff and open Pending Requests.
Test Data	Blood group: A+; Units: 2; Reason: Surgery
Expected Result	Request is created with a unique ID and status Pending. Hospital Staff sees it in My Requests with status Pending. Blood Bank Staff sees it in the Pending Requests list.
Actual Result	
Status	Not Executed

TC-REQ-02	Reject a blood request with missing mandatory data
Requirement IDs	FR-6.7, FR-6.2
Type / Priority	Functional (Negative) / High
Preconditions	Logged in as Hospital Staff.
Test Steps	1. Open New Blood Request. 2. Select A+ but leave Units empty. 3. Submit.
Test Data	Blood group: A+; Units: (blank)
Expected Result	Request is NOT created. Validation message states that Units is required. No new record in the requests list.

Actual Result	
Status	Not Executed

TC-ISS-01	Approve a request and issue blood with sufficient stock
Requirement IDs	FR-7.1, FR-7.5, FR-7.6, FR-7.7, FR-3.3, SAFE-04, BR-06, BR-07
Type / Priority	Functional / Safety / High
Preconditions	Pending request R1 for 2 units A+. A+ stock = 6. Logged in as Blood Bank Staff.
Test Steps	1. Open R1 and click Approve. 2. Click Issue Blood. 3. At the confirmation prompt click Cancel; check stock. 4. Click Issue Blood again and Confirm.
Test Data	Request R1: A+, 2 units; A+ stock: 6
Expected Result	R1 status becomes Approved. Cancelling the confirmation leaves stock at 6. After confirming, stock = 4, R1 status = Fulfilled, and an issue record linked to R1 is created.
Actual Result	
Status	Not Executed

TC-ISS-02	Block issue against a pending or rejected request
Requirement IDs	FR-7.1, FR-7.2, SAFE-02, BR-04
Type / Priority	Functional / Safety (Negative) / High
Preconditions	Request R2 is Pending; Request R3 has been Rejected by staff.
Test Steps	1. Attempt to issue blood for R2 (UI and direct POST to /api/issues). 2. Attempt the same for R3.
Test Data	R2: Pending; R3: Rejected
Expected Result	Issue is refused for both with a message that the request must be approved first. Inventory is unchanged; no issue record is created.
Actual Result	
Status	Not Executed

TC-ISS-03	Block issue when stock is insufficient
Requirement IDs	FR-7.3, FR-7.4, FR-3.5, SAFE-03, BR-05
Type / Priority	Functional / Safety (Negative) / High

Preconditions	Approved request R4 for 3 units B+. B+ stock = 2.
Test Steps	1. Open R4. 2. Click Issue Blood and Confirm.
Test Data	R4: B+, 3 units; B+ stock: 2
Expected Result	Issue is refused with an 'insufficient stock' message. B+ stock stays 2 (never negative). R4 stays Approved.
Actual Result	
Status	Not Executed

TC-RPT-01	View history and basic reports
Requirement IDs	FR-8.1-FR-8.5
Type / Priority	Functional / Medium
Preconditions	Data from earlier test cases exists. Logged in as Administrator.
Test Steps	1. Open Reports & History. 2. View Donation, Request, Issue and Inventory reports one by one.
Test Data	Records created in TC-DNT-01, TC-REQ-01, TC-ISS-01
Expected Result	Each report lists the expected records with dates and statuses matching the database (e.g., R1 shows Fulfilled, 2 units A+).
Actual Result	
Status	Not Executed

TC-PERF-01	Search and inventory view response time
Requirement IDs	NFR-01
Type / Priority	Non-functional (Performance) / High
Preconditions	Test DB seeded with 10,000 donors and 30,000 units. Load tool configured for 20 concurrent users.
Test Steps	1. Run a 10-minute load test of donor search, availability search and inventory view with 20 concurrent users. 2. Record response times.
Test Data	Locust script; 20 users; mixed queries across all 8 blood groups
Expected Result	95th-percentile response time $\leq$ 3 s for each operation; 0% server errors.
Actual Result	
Status	Not Executed

TC-PERF-02	Transaction submission response time
Requirement IDs	NFR-02
Type / Priority	Non-functional (Performance) / Medium
Preconditions	Same environment as TC-PERF-01.
Test Steps	1. Under 20 concurrent users, submit donations, blood requests and issues. 2. Record response times.
Test Data	Locust script; valid payloads
Expected Result	95th-percentile processing time $\leq$ 3 s for each submission type.
Actual Result	
Status	Not Executed

TC-PERF-03	Report generation time
Requirement IDs	NFR-03
Type / Priority	Non-functional (Performance) / Medium
Preconditions	Same seeded DB as TC-PERF-01.
Test Steps	1. Open each report type 10 times. 2. Record page load time.
Test Data	10,000 donors; 30,000 units; 5,000 requests
Expected Result	Every report renders within 5 s.
Actual Result	
Status	Not Executed

TC-REL-01	Failed issue does not partially update inventory
Requirement IDs	NFR-06
Type / Priority	Non-functional (Reliability) / High
Preconditions	Approved request R5 for 2 units O-. O- stock = 5. A fault is injected so that saving the issue record raises a database error.
Test Steps	1. Issue blood for R5 with the fault active. 2. Check O- stock, R5 status and issue records.
Test Data	Fault injection via test mock on IssueRecord.save()

Expected Result	Error message is shown. O- stock remains 5, R5 remains Approved, and no issue record exists (transaction rolled back).
Actual Result	
Status	Not Executed

TC-USE-01	Staff can record a donation within 2 minutes
Requirement IDs	NFR-05
Type / Priority	Non-functional (Usability) / Medium
Preconditions	3 testers given a 5-minute walkthrough of the system.
Test Steps	1. Each tester records a donation for an existing donor from the dashboard. 2. Time each attempt.
Test Data	Pre-registered donor; stopwatch
Expected Result	Each tester completes the task within 2 minutes without assistance.
Actual Result	
Status	Not Executed

TC-PORT-01	Cross-browser compatibility
Requirement IDs	NFR-08
Type / Priority	Non-functional (Portability) / Low
Preconditions	Latest Chrome, Edge and Firefox on Windows, macOS and Linux.
Test Steps	1. Run the smoke suite (login, search, request, issue) on each browser/OS pair.
Test Data	Selenium smoke suite
Expected Result	All smoke tests pass on every browser/OS pair; no layout breaks on key screens.
Actual Result	
Status	Not Executed

TC-SEC-01	Unauthenticated access is blocked
Requirement IDs	SEC-01, SO-01
Type / Priority	Security / High
Preconditions	Browser session not logged in.
Test Steps	1. Directly open /inventory/, /requests/ and /reports/ URLs. 2. Send the same requests via API client without a session.
Test Data	Protected URLs list
Expected Result	Without logging in, send requests to GET /api/inventory, GET /api/requests and GET /api/reports. 2) Open the corresponding pages in the browser.
Actual Result	
Status	Not Executed

TC-SEC-02	Role-based access: Hospital Staff cannot modify inventory or issue blood
Requirement IDs	SEC-02, SEC-03, SO-02, SO-03
Type / Priority	Security / High
Preconditions	Logged in as Hospital Staff.
Test Steps	Check the menu for donation, approval and issue options. 2) Send direct POST requests to /api/donations, /api/requests/{id}/approve and /api/issues.
Test Data	Hospital Staff account
Expected Result	Options are not shown. All three POSTs return 403 Forbidden. Inventory and request statuses are unchanged. The rejected attempts are recorded in the application log.
Actual Result	
Status	Not Executed

TC-SEC-03	Only Administrator can manage users
Requirement IDs	SEC-04, SO-03
Type / Priority	Security / High
Preconditions	Accounts exist for Staff, Hospital Staff and Administrator.
Test Steps	1. As Staff and as Hospital Staff, send GET and POST to /api/users. 2. Repeat as Administrator.
Test Data	Three role accounts
Expected Result	Staff and Hospital Staff receive 403. Administrator can create the user successfully.

Actual Result	
Status	Not Executed

TC-SEC-04	Password masking and secure storage
Requirement IDs	SEC-05, SO-01
Type / Priority	Security / High
Preconditions	Login and user-creation screens available. DB access for tester.
Test Steps	1. Type a password on the login screen. 2. Create a user and inspect the stored password field in the DB.
Test Data	Password: Test@1234
Expected Result	Password characters are masked on screen. The DB stores only a salted hash (e.g., pbkdf2_sha256\$...), never plain text.
Actual Result	
Status	Not Executed

TC-SEC-05	HTTPS enforced on deployment
Requirement IDs	SEC-06
Type / Priority	Security / Medium
Preconditions	Application deployed on the staging server with TLS certificate.
Test Steps	1. Open the site using http://. 2. Inspect response headers and cookies.
Test Data	Staging URL
Expected Result	HTTP is redirected to HTTPS. Session and CSRF cookies have the Secure flag. TLS 1.2 or higher is negotiated.
Actual Result	
Status	Not Executed

TC-SEC-06	Injection and XSS attempts in search and request fields
Requirement IDs	SO-01, SO-02
Type / Priority	Security / High
Preconditions	Logged in as Hospital Staff.
Test Steps	1. Enter SQL-injection payloads in search and request fields. 2. Enter script payloads in the request Reason field and view the request.
Test Data	' OR '1'='1 ; <script>alert(1)</script>
Expected Result	No extra data is returned and no DB error is shown. Script is displayed as plain text and never executes.
Actual Result	
Status	Not Executed

16. Approvals

Role	Name	Signature / Date
QA Lead	Purav N Khincha (PES1UG24CS349)	
Requirements & SRS Lead	Prakruti Prasanna Bhat (PES1UG24CS330)	
Architecture & Design Lead	Pallavi Rajkumar Bubanale (PES1UG24CS314)	
Project Lead / Product Owner	Ramith Munnar Ravindranath (PES1UG24CS367)	